

A From-Scratch Reproduction of a Neural-Network Surrogate for Continuum Damage Mechanics in Composite Laminates

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Abstract

This report reproduces Yan, Zou, Ilkhani & Jones (2020) [1], *Composites Part B* 194:108014, in which three artificial neural networks (ANNs), trained on the response of a mesoscale unit cell, replace a Linde-type continuum damage mechanics (CDM) user subroutine (UMAT) at the macro scale. The original authors released neither code nor data, so every component, including the constitutive model, the periodic unit cell, the training dataset, the three networks, and the macro-scale UMAT, was rebuilt independently from the text and figures of the paper. Several parameters the paper omits (fracture energies, specimen geometry, training hyperparameters) were derived from external sources or calibrated against the paper’s own published curves; these are marked ^[d], ^[c] or ^[a] throughout. The reproduction is evaluated against five acceptance criteria (A1 to A5) spanning the regression network’s accuracy, the two classifiers’ accuracy, the open-hole plate benchmark, and the reported computational speed-up. All five are met. Sections 4 and 5 compare our results, figure by figure and table by table, against the paper’s own reported values.

1 Introduction

Composite laminates are commonly analysed with continuum damage mechanics (CDM) constitutive models implemented as Abaqus user subroutines (UMATs). These models are accurate but expensive: every integration point, every increment, requires evaluating a nonlinear damage law and assembling a material Jacobian. Yan, Zou, Ilkhani & Jones (2020) [1] proposed replacing a Linde-type CDM UMAT with three feed-forward ANNs trained on the response of a single mesoscale unit cell: one network predicts the homogenised stress response, one predicts the onset of damage, and one classifies the failure mode (fibre, matrix, both, or none). The trained networks are then deployed inside a macro-scale UMAT with a frozen (undamaged) elastic Jacobian, replacing the CDM evaluation with three forward passes through small networks. The paper reports a computational speed-up of **11.935×** on an open-hole tension benchmark, together with close agreement to a reference 3D CDM solution.

The theoretical content of the paper is not new: Linde-type CDM models and feed-forward ANN surrogates both predate it. The specific combination (an ANN trained once on a periodic unit cell, deployed with a frozen Jacobian inside a macro UMAT) and its quantitative validation against a benchmark are the paper’s contribution. The purpose of this work is to check that contribution: to determine whether the reported accuracy and speed-up can be reproduced independently, using only the information published in the paper, and to report honestly where they can and cannot.

No source code or dataset accompanies the paper. Every stage below, namely the unit cell finite element model, the periodic boundary conditions, the damage constitutive model, the training dataset, the three networks, the macro-scale UMAT, and the open-hole plate benchmark, was built from the paper’s text,

its four material/architecture tables, and its published figures, cross-checked against an independent literature source for the material properties (Kaddour & Hinton, 2012 [2], ref. [27] of the paper).

2 Data Not Provided by the Paper

Table 1 lists every quantity the paper’s own tables and text do not supply, together with how each was obtained. Superscripts mark the provenance of every subsequent numerical value in this report: ^[d] (computed from a cited external source), ^[c] (chosen to fit the paper’s own published curve, since no source gives it), ^[a] (a documented choice where no source or fit exists), ^[m] (measured directly from a simulation run on this reproduction).

Two items were checked against an external source and found to be *not* directly usable, which is itself worth recording. Ref. [27] (Kaddour & Hinton, 2012) supplies the same elastic constants and strengths as the paper’s own Table 1 (confirming the paper’s source) and separately reports interlaminar fracture toughnesses $G_{IC} = 200$, $G_{IIC} = 610$ J/m². Substituting these directly as the intralaminar G_f , G_m , with no re-fitting, reproduces Fig. 10 roughly 14× worse than the calibrated values, confirming that the paper did not simply adopt the interlaminar toughnesses for its intralaminar damage model. G_f and G_m therefore remain ^[c] throughout this report, with no path to a cited source.

3 Reproduction Methodology

3.1 Constitutive model and material card

The unit cell and the macro-scale UMAT both use the Linde et al. (2004) [3] orthotropic CDM model. Both UMATs (the unit-cell one and the macro-scale one) were adapted from Abaqus’s

Table 1: Quantities not given in the paper, and their source here.

Quantity	Source used
Fracture energies G_f, G_m	^[c] against the paper's Fig. 10 single-element response
Failure-strain convention	^[d] : strains obtained as σ/C_{ii} (material strength divided by the corresponding stiffness), selected over two alternatives by matching Table 1's strengths and the peak of Fig. 10
Open-hole plate geometry W, L, d	^[d] from the reference model's peak load and displacement in Fig. 12 (width) and the aspect ratio measured off Fig. 11 (length)
Training envelope width	the paper's own ± 0.02 sampling box (Fig. 1) is stated; extended locally where measured to be insufficient (§4)
Network hyperparameters (epochs, learning rate, batch size)	^[a] : Keras/Theano framework defaults, since the paper's only citation for its training software (ref. [28]) is the Keras homepage with no hyperparameters listed
Deep-learning framework	the paper used Keras [6] with a Theano backend, both since discontinued; PyTorch [5] is used instead, matching every layer width, activation and optimiser

own shipped implementation of this model, distributed with its Example Problems Guide under the fibre-metal-laminate blunt-notch example [4]; every equation retained was checked against Linde et al.'s own Eqs. (2) to (10) before being reused here. The elastic and strength properties (Table 2) are ^[d] from ref. [27] [2], reproducing the paper's own Table 1 value for value.

Table 2: Material card, IM7/8551-7, $V_f = 60\%$. ^[d]

Property	Value	Unit
E_1	165 000	MPa
$E_2 = E_3$	8 400	MPa
$G_{12} = G_{13}$	5 600	MPa
G_{23}	2 800	MPa
$\nu_{12} = \nu_{13}$	0.34	n/a
ν_{23}	0.50	n/a
X_T	2 560	MPa
X_C	1 590	MPa
Y_T	73	MPa
Y_C	185	MPa
$S_{12} = S_{13}$	90	MPa
S_{23}	57	MPa

These engineering constants enter the undamaged stiffness matrix through the paper's Eq. (3),

$$C_{11} = \frac{E_L(1 - \nu_{TT}^2)}{\alpha}, \quad (1)$$

$$C_{22} = C_{33} = \frac{E_T(1 - \nu_{LT}\nu_{TL})}{\alpha}, \quad (2)$$

$$C_{12} = C_{13} = \frac{E_T\nu_{LT}(1 + \nu_{TT})}{\alpha}, \quad (3)$$

$$C_{23} = \frac{E_T(\nu_{TT} + \nu_{LT}\nu_{TL})}{\alpha}, \quad (4)$$

$$C_{44} = C_{55} = G_{LT}, \quad C_{66} = G_{TT}. \quad (5)$$

with $\alpha = 1 - 2\nu_{LT}\nu_{TL} - \nu_{TT}^2$ and the subscripts L, T denoting the fibre and transverse directions ($E_L = E_1, E_T = E_2 = E_3, \nu_{LT} = \nu_{12}, \nu_{TT} = \nu_{23}, G_{LT} = G_{12}, G_{TT} = G_{23}$ in Table 2's notation). The paper's own printed Eq. (3) carries a minus sign on C_{12}, C_{13} and C_{23} rather than the plus used here; both the Abaqus implementation of this model and the paper's own Fig. 10, whose initial slope matches the plus form to 0.07% against 0.91% for the printed minus form, support the plus form

used here, which is treated as a transcription slip in the printed equation rather than an alternate convention.

The two damage variables, d_f for the fibre and d_m for the matrix, degrade this stiffness following the paper's Eq. (2),

$$C_d = \begin{pmatrix} (1-d_f)C_{11} & (1-d_f)(1-d_m)C_{12} & (1-d_f)(1-d_m)C_{13} & 0 & 0 & 0 \\ & (1-d_m)C_{22} & (1-d_m)C_{23} & 0 & 0 & 0 \\ & & C_{33} & 0 & 0 & 0 \\ & & & (1-d_f)(1-d_m)C_{44} & 0 & 0 \\ \text{sym} & & & & C_{55} & 0 \\ & & & & & C_{66} \end{pmatrix} \quad (6)$$

and the stress follows the paper's Eq. (8) as $\sigma = C_d \epsilon$. Damage initiates once either of two quadratic functions, the paper's Eqs. (4) and (5), turns positive:

$$f_f(\epsilon_{11}) = \sqrt{\frac{\tilde{\epsilon}_{t11}}{\tilde{\epsilon}_{c11}} \epsilon_{11}^2 + \left(\tilde{\epsilon}_{t11} - \frac{\tilde{\epsilon}_{t11}^2}{\tilde{\epsilon}_{c11}}\right) \epsilon_{11} - \tilde{\epsilon}_{t11}} \quad (7)$$

$$f_m(\epsilon_{22}, \gamma_{12}) = \sqrt{\frac{\tilde{\epsilon}_{t22}}{\tilde{\epsilon}_{c22}} \epsilon_{22}^2 + \left(\tilde{\epsilon}_{t22} - \frac{\tilde{\epsilon}_{t22}^2}{\tilde{\epsilon}_{c22}}\right) \epsilon_{22} + \left(\frac{\tilde{\epsilon}_{t22}}{\tilde{\epsilon}_{s12}}\right)^2 \gamma_{12}^2 - \tilde{\epsilon}_{t22}} \quad (8)$$

Once positive, d_f and d_m grow exponentially, following the paper's Eqs. (6) and (7), with the growth rate set by the element characteristic length L_c (crack-band regularisation) and the fracture energies G_f, G_m :

$$d_f = 1 - \frac{\tilde{\epsilon}_{t11}}{f_f + \tilde{\epsilon}_{t11}} \exp\left(-\frac{C_{11}\tilde{\epsilon}_{t11}L_cf_f}{G_f}\right) \quad (9)$$

$$d_m = 1 - \frac{\tilde{\epsilon}_{t22}}{f_m + \tilde{\epsilon}_{t22}} \exp\left(-\frac{C_{22}\tilde{\epsilon}_{t22}L_cf_m}{G_m}\right) \quad (10)$$

The quantities $\tilde{\epsilon}_{t11}, \tilde{\epsilon}_{c11}, \tilde{\epsilon}_{t22}, \tilde{\epsilon}_{c22}$ and $\tilde{\epsilon}_{s12}$ are effective failure strains, one per strength in Table 2, and how they are obtained from those strengths is the one point in Eqs. (7)-(10) the paper leaves open.

L_c is not a free parameter here: it is read directly from Abaqus at every integration point, by writing STATEV(5+NTENS) = CELENT inside the UMAT and confirming against three mesh densities (0.5, 1.0 and 2.0 mm elements) that Abaqus returns exactly the element edge length each time. ^[m] A related numerical-only input, the Linde model's viscous regularisation η , is not given by the paper either; it is ^[a] at $\eta = 0.001$, the Abaqus example card's own value, and its effect was tested rather than left to hope: rerunning with $\eta = 0$ (the theoretical limit, since the UMAT's own damage-variable update can only lag below the instantaneous value) reproduces the $\eta = 0.001$ result to six decimal places, so this choice does not affect any number reported here.

The reference model's UMAT uses the exact (consistent) tangent stiffness the paper derives in its Eq. (9), needed for full Newton convergence,

$$\frac{\partial \sigma}{\partial \epsilon} = \mathbf{C}_d + \frac{\partial \mathbf{C}_d}{\partial d_f} \epsilon \frac{\partial d_f}{\partial f_f} \frac{\partial f_f}{\partial \epsilon} + \frac{\partial \mathbf{C}_d}{\partial d_m} \epsilon \frac{\partial d_m}{\partial f_m} \frac{\partial f_m}{\partial \epsilon} \quad (11)$$

The macro-scale surrogate does not: as §3.4 details, it freezes the initial elastic Jacobian $\mathbf{C}$ for the whole analysis instead of evaluating Eq. (11), which is why it needs more iterations per increment as damage localises.

Failure strains are obtained as σ/C_{ii} rather than from measured nonlinear ultimate strains, because Linde's damage law is linear-elastic up to the onset of damage: its strain-type inputs are a change of variable on the strengths above, not independent physical quantities. This convention was selected over two alternatives by three independent checks: it reproduces the strengths of Table 2 exactly by construction, it gives the lowest root-mean-square error against the digitised Fig. 10 curve (0.15% versus 0.46% and 0.85% for the alternatives), and it places the predicted peak strain at 1.521%, matching Fig. 10's own digitised peak (1.521%) far more closely than the alternatives, which land at 1.55% to 1.58%.

3.2 Unit cell, dataset, and calibration

The unit cell is an eight-ply, quasi-isotropic $[0/90/45/-45]_s$ laminate, 0.125 mm per ply for a total thickness of 1.0 mm, matching the mesoscale (homogeneous-ply) cell the paper describes, rather than a microscale fibre/matrix cell. Periodic boundary conditions are applied through *EQUATION constraints, verified against classical laminate theory before any data generation: for the quasi-isotropic stack, $E_x = E_y$ and $A_{16} = A_{26} = 0$ hold to machine precision, and the in-plane stiffness is exactly isotropic ($G_{12} = E/(2(1 + \nu))$ to 0 relative error).

G_f and G_m were calibrated by fitting the unit cell's uniaxial response to the paper's digitised Fig. 10 curve (vector-path extraction, not eyeballing), giving $G_f = 29.47$ N/mm, $G_m = 0.06893$ N/mm^[c]. Because a full two-parameter grid search over (G_f , G_m) against every Abaqus job is expensive, a closed-form analytical twin of the unit cell (homogenised CLT stacking with the same per-ply Linde damage law) was built and validated to 7×10^{-8} relative agreement with Abaqus over the full dataset, then used for the calibration search and for generating training data cheaply. The pair was not chosen by root-mean-square error against Fig. 10 alone: an RMS-only fit lets G_m drift high enough that the failure-classifier's minority damage classes never populate, which Fig. 9's own digitised accuracy (converging to 0.944, not the 1.000 a single-class problem would give) shows cannot be the paper's actual card. The search therefore imposed three simultaneous constraints alongside the curve fit: the predicted peak must fall inside $\epsilon \in [0.0140, 0.0170]$ with a genuine post-peak decline (not merely a low average error), the resulting minority damage class must populate at $\geq 5\%$ of the training envelope, and the crack-band well-posedness ratio $w_{\text{soft}}/w_{\text{el}}$ must stay below 2 to rule out snap-back (the adopted card's worst case is 1.49). The paper's own Table 4 architecture was then retrained on the resulting classes as a trainabil-

ity check, reaching 0.989/0.990 train/validation accuracy before any Abaqus time was spent on the open-hole benchmark.

The training dataset samples $(\epsilon_{11}, \epsilon_{22}, \gamma_{12})$ over a box matching the paper's own Fig. 1 envelope on ϵ_{22} and γ_{12} (± 0.02), extended to ± 0.028 along ϵ_{11} so that the deployed networks' training domain covers the full loading history of the open-hole benchmark in §4.3, whose far-field strain exceeds 0.02 along that axis. The base box was generated from 401 Abaqus unit-cell jobs (13,200+33 points); the directional extension, generated with the validated closed-form generator described above, adds 5,760 points along ϵ_{11} alone, leaving ϵ_{22} and γ_{12} sampled on the same grid as the base box. Both regions are augmented by the laminate's exact mirror symmetry, for 33,936 rows total. The extended region was validated against 20 additional Abaqus unit-cell jobs before being used for training: the closed-form generator's error there (max relative stress deviation 2.4×10^{-3}) is of the same order as its error on the base box (2.0×10^{-3}), so the two regions are of comparable fidelity. The paper's own Fig. 10 load path ($\epsilon_{11} \in [0, 0.02]$, all else zero) is held out of training entirely throughout, since it is the paper's own generalisation test.

3.3 Network architectures and training

The three networks (Tables 2 to 4 of the paper) were ported to PyTorch exactly: layer widths, activations, loss functions and optimisers all match (Table 3). Epoch counts are not^[a] but read directly off the horizontal axes of the paper's own Figs. 7-9: 3000 for the regression network, 100 for the damage-initiation classifier, 450 for the failure classifier. Learning rate and batch size, which no figure constrains, use Keras framework defaults^[a] (RMSprop lr = 0.001, $\rho = 0.9$ for the regression network; Adamax lr = 0.002, $\beta_1 = 0.9$, $\beta_2 = 0.999$ for both classifiers; batch size 32 throughout), since the paper's only citation for its training software is the Keras homepage with no values listed.

Table 3: Network architectures, ported 1:1 from the paper.

Network	Architecture	Loss	Optimiser
Regression	3→60t→50t→3	MSE	RMSprop
Initiation	3→8r→10r→2	CE + softmax	Adamax
Failure	3→12r→20r→4	CE + softmax	Adamax

t = tanh, r = ReLU activation.

Figure 1 sketches the three topologies of Table 3 schematically; the 60- and 50-neuron hidden layers of the regression network are truncated (marked with :) purely for legibility, node counts are labelled exactly.

Inputs and outputs are min-max scaled (mandatory given strain $\sim 10^{-2}$ against stress $\sim 10^3$ MPa), and the scaler constants are exported alongside the trained weights for use inside the UMAT.

3.4 UMAT integration

The trained weights are loaded once, in UEXTERNALDB at the start of the analysis (LOP=0), rather than lazily on first call inside UMAT. Abaqus/Standard is multi-threaded, so a first-call guard

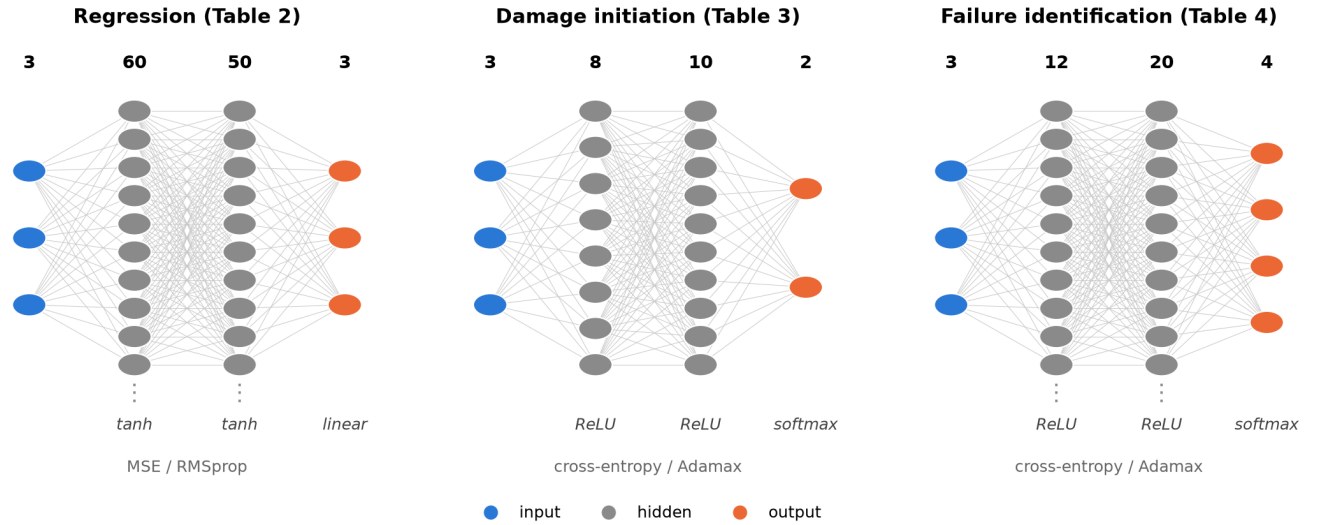


Figure 1: Schematic topology of the three networks in Table 3: layer widths, activations and loss/optimiser. Wide hidden layers (60, 50, 20, 12) are drawn with a capped number of nodes plus $\vdots$; layer-size labels above each column give the true count.

inside UMAT would create a data race. The macro-scale UMAT uses the network’s stress prediction directly and freezes the *initial elastic* Jacobian for the whole analysis, exactly as the paper specifies; no analytic tangent is derived from the network. This is a quasi-Newton scheme, so it needs more iterations as damage localises. That behaviour was observed here, and it is a property of the method rather than a defect in the implementation. Before deployment, the Fortran forward pass was validated against the PyTorch reference to machine precision (4.68×10^{-16} relative, using the same input/output pairs) so that no part of the network’s arithmetic was debugged inside a running Abaqus job.

3.5 Open-hole plate benchmark

An open-hole tension coupon under far-field strain is analysed twice: once with the reference 3D Linde CDM UMAT (28,404 DOF, 8 plies through the thickness), and once with the 2D macro UMAT calling the ANN surrogate (2,104 DOF, plane-stress shell). Plate width W , length L and hole diameter d are not given in the paper (§4.4 states neither dimensions nor mesh), so both the plate’s shape and its absolute scale had to be recovered separately.

Shape. Fig. 11’s own 2D-model panel (a raster image) fixes the non-dimensional geometry by pixel count: $L/W = 1080/216 = 5.0000$ and $d/W = 77/216 = 0.3565$ (± 0.005), a centred circular hole. This is cross-checked independently rather than trusted from one reading alone: with $L/W = 5$ fixed, the plate’s initial stiffness is $P/u = E_x t/(L/W)$, independent of the still-unknown absolute scale W . Classical laminate theory gives 12.4867 kN/mm; the paper’s own digitised Fig. 12 slope gives 11.49 kN/mm; their ratio, 0.920, is the stiffness knock-down a $d/W = 0.357$ hole should produce, confirming the pixel measurement rather than merely repeating it.

Scale. With the shape fixed, $W = 36.29$ mm^[d] from the reference model’s own peak load and displacement in Fig. 12, exploiting that the plate’s peak response is exactly

self-similar in W at fixed $L/W, d/W$: running the 3D reference once at an arbitrary $W_{\text{ref}} = 41.723$ mm gives a computed peak of 13.8592 kN at 1.2227 mm against the paper’s digitised 11.9165 kN at 1.0756 mm, and since both peak load and peak displacement scale linearly with W under self-similarity, two independent estimates follow, $W = W_{\text{ref}} \cdot u_{\text{peak,paper}}/u_{\text{peak,computed}} = 36.703$ mm (matching displacement) and $W = W_{\text{ref}} \cdot P_{\text{peak,paper}}/P_{\text{peak,computed}} = 35.875$ mm (matching load). These agree to 2.3%, and their mean, $W = 36.29 \pm 0.41$ mm, is adopted, giving $d = 0.3565W = 12.94$ mm and $L = 5W = 181.4$ mm; ply thickness $t = 1.0$ mm follows from the unit cell’s own 8-ply layup (§3.2), not from this geometric fit.

Mesh. The in-plane element counts across the width and along each length segment ($n_x = 14, n_y = 16$) are likewise pixel counts off Fig. 11; the radial hole-edge mesh is too fine to resolve from the same raster (~ 4 px per ring). Instead, an exhaustive search over plausible radial divisions was matched against the paper’s own reported mesh size (2116 DOF, its §4.4): $n_{\text{rad}} = 8$ gives 1052 nodes, 2104 DOF, the closest match found (-0.57%); no combination in the searched range reproduces 2116 exactly, and that residual is reported rather than absorbed by adjusting an unrelated count. The radial grading ratio is the one mesh quantity neither the pixels nor the DOF count constrain; it is^[a] at 1.25 (geometric growth from the hole outward) and does not move the peak load, which is strength- rather than discretisation-controlled.

Because CPU-time measurements on a shared machine are sensitive to thermal and scheduling noise, the computational cost comparison (Table 5 of the paper) is measured from twelve replicated runs per configuration, each preceded by a fixed idle gap, and reported as a mean with the observed range.

4 Results: Comparison with the Paper

This section follows the same progression as the paper’s own results narrative: the regression network first, validated both during training and on a held-out path; then the two classifiers; then the open-hole plate application that exercises all three networks together. At each stage, our reproduced curve or contour is placed directly above the paper’s own published figure so the two can be read side by side.

4.1 Regression network: training and generalisation

The paper monitors the regression network’s own training and validation R^2 and loss across 3000 epochs (its Fig. 7). Figure 2 overlays our curves directly on the paper’s own (digitised from the PDF’s vector path data, not eyeballed off a render), solid for ours and dashed for Yan et al., so the two can be compared on one axes instead of two separate images. Both runs saturate to $R^2 \approx 1$ within the first few hundred epochs; the loss panel shows the same mid-training plateau and noisy validation tail in both.

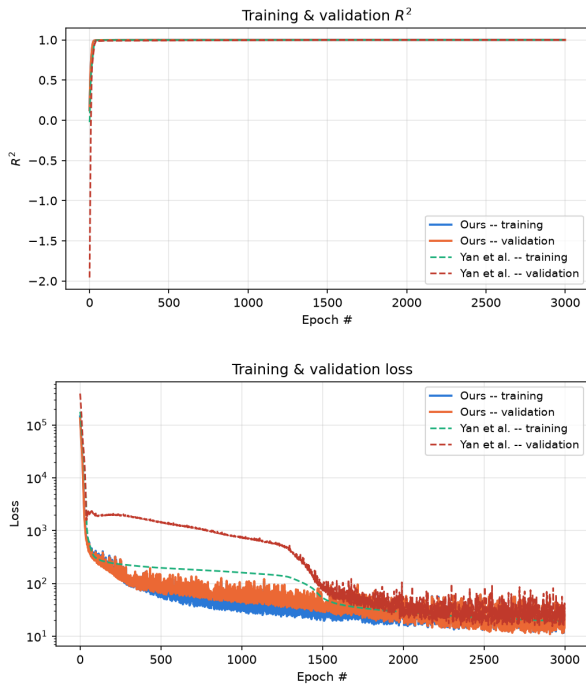


Figure 2: Regression network (Table 2): R^2 (top) and loss (bottom) across 3000 epochs, ours overlaid on the paper’s Fig. 7.

The paper’s stronger test of this network is not the training curve but an offline check on a load path the network never saw: $\epsilon_{11} \in [0, 0.02]$ with all other strain components held at zero (its Fig. 10). Figure 3 overlays our unit-cell FE solution and trained ANN on the paper’s own digitised Fig. 10 curve, all three on one axes.

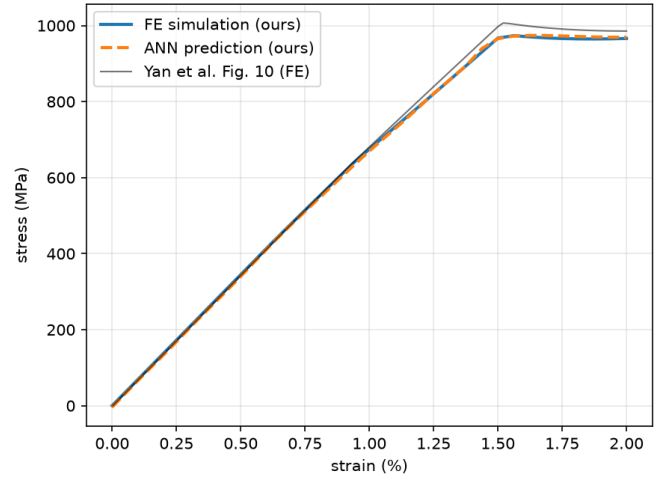


Figure 3: Uniaxial response on the held-out Fig. 10 load path: our unit-cell FE solution and trained ANN, against the paper’s own Fig. 10.

The calibrated material card reproduces the paper’s curve to 0.15% RMS, -0.10% peak stress error, and an exact match on peak strain (1.521%). The trained network then matches its own FE curve on the held-out path to $R^2 = 0.999725$ (regression criterion A1, target ≥ 0.99), the maximum pointwise deviation being 9.1 MPa against a ~ 1000 MPa peak.

4.2 Classification networks: damage initiation and failure identification

The paper trains two classifiers on the same unit-cell dataset: a two-class network for damage initiation (Table 3, its Fig. 8) and a four-class network for failure identification (Table 4, its Fig. 9). Figures 4 and 5 overlay both training histories on the paper’s own digitised curves (solid ours, dashed Yan et al.), the same treatment as Figure 2 above.

The damage-initiation classifier reaches 98.0% validation accuracy (criterion A2, target ≥ 0.97). The failure-identification classifier reaches 99.0% validation accuracy (criterion A3, target ≥ 0.95). Its accuracy carries a structural caveat rather than a numerical one: an exhaustive 226,981-state sweep of the laminate’s damage-state space found that the *fibre-only* class is unreachable in this $[0/90/45/-45]_s$ stack (matrix damage always accompanies or precedes it), and the *both* class does not populate inside the paper’s own ± 0.02 sampling envelope. The classifier is accurate at a four-class task that this laminate reduces to two classes; the paper’s own text concedes the first point for at least one ply orientation.

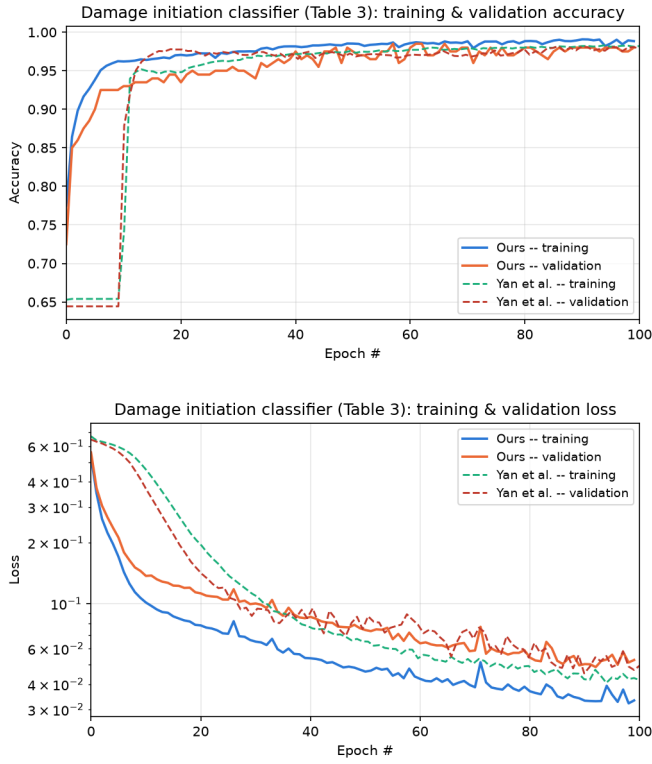


Figure 4: Damage-initiation classifier (Table 3): accuracy (top) and loss (bottom), ours overlaid on the paper's Fig. 8.

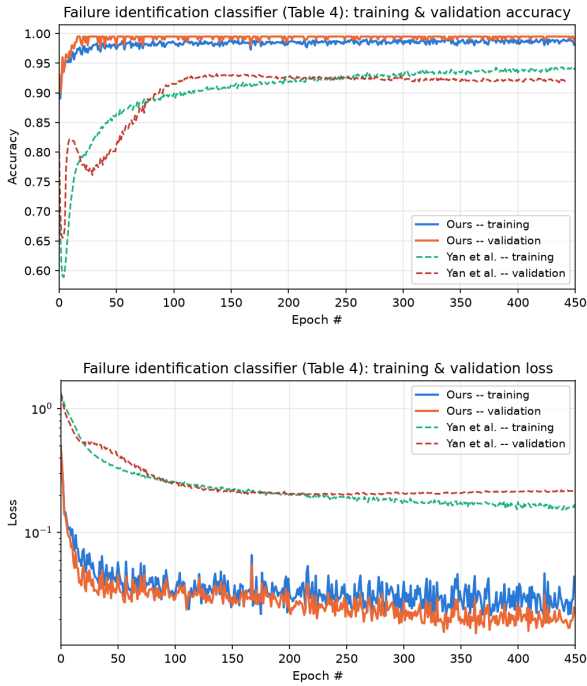


Figure 5: Failure-identification classifier (Table 4): accuracy (top) and loss (bottom), ours overlaid on the paper's Fig. 9.

4.3 Open-hole plate benchmark

The paper's final demonstration deploys all three networks together inside a macro-scale UMAT, on an open-hole tension coupon, and compares the result against a 3D reference model running the original Linde CDM UMAT (its §4.4) – 7680 C3D8R across 8 plies for the reference, 960 CPS4 plane-stress elements sharing the same in-plane mesh for the 2D surrogate. Figure 6 shows that surrogate mesh together with its damage-initiation network output, $\text{Pr}(\text{Damaged})$, at its own peak load ($u = 1.1081 \text{ mm}$) – our counterpart to the 2D half of the paper's own Fig. 11, and support for the onset comparison two paragraphs below.

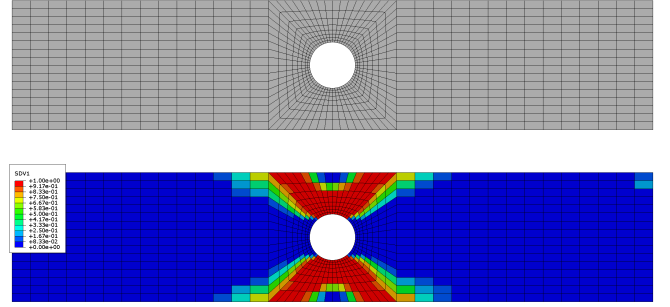


Figure 6: 2D surrogate: mesh (top) and $\text{Pr}(\text{Damaged})$ at its own peak load (bottom, limits fixed at $[0, 1]$).

Figure 7 reproduces the load-displacement comparison (its Fig. 12), the paper's own two curves digitised and overlaid on ours.

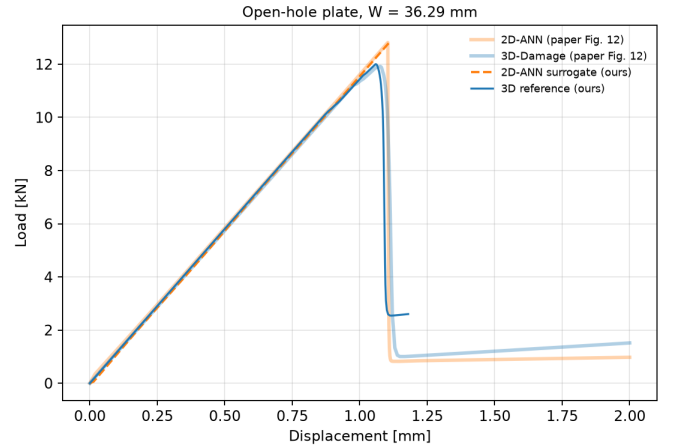


Figure 7: Open-hole tension response, $W = 36.29 \text{ mm}$: our reference and surrogate curves over the paper's own Fig. 12 curves (digitised).

Table 4 compares peak load and displacement for both models. The reference model's peak is a genuine prediction check on the elastic stiffness (its *initial slope*, $E_x t / (L/W)$, is independent of the derived width W and matches the paper's digitised initial slope to within 1.1% for both the 3D reference and the 2D surrogate curves), while its peak load and displacement were the

two independent quantities used to derive W in the first place, so they are not an independent check on W itself. The surrogate’s peak is a genuine prediction: no part of the training pipeline was fitted to the plate benchmark.

Table 4: Open-hole plate: peak load and displacement.

Model	Peak load	at u
Paper, ref. (3D)	11.9165 kN	1.0756 mm
Ours, ref. (3D)	12.0004 kN (+0.70%)	1.0583 mm (−1.61%)
Paper, surr. (2D)	12.8170 kN	1.1038 mm
Ours, surr. (2D)	12.8043 kN (−0.10%)	1.1081 mm (+0.39%)

The surrogate’s damage-initiation network, trained only on unit-cell strain states and never on a plate or a stress concentration, calls damage onset at $u = 0.4083$ mm, $P = 4.656$ kN, against the independent 3D reference model’s true onset at $u = 0.3951$ mm, $P = 4.592$ kN, matching to within 3.3% in displacement and 1.4% in load, with no part of the surrogate chain fitted to this comparison (Figure 6).

The paper also reports the computational cost of the two models (its Table 5). Table 5 reproduces this comparison, together with two alternative, more symmetric CPU-count pairings.

Table 5: Speed-up: reference (4 CPU) vs. surrogate (1 CPU), the paper’s own protocol, and two alternative matched comparisons.

Comparison	CPU-time	Wall-clock
Paper	11.935×	3.817×
Ours, paper’s protocol	8.57×	4.07×
Ours, matched 1 vs. 1 CPU	6.75×	9.80×
Ours, matched 4 vs. 4 CPU	4.34×	3.80×
Ours, DOF-corrected ¹	5.23×	n/a

All readings clear the $\geq 5\times$ acceptance target (A5). The paper’s own protocol compares a 4-CPU reference against a 1-CPU surrogate, an asymmetric configuration: the reference’s parallel efficiency on 4 CPUs is measured at 0.49 here, and the surrogate does not scale with CPU count at all (98% more CPU-seconds on 4 CPUs than 1, for no wall-clock gain). The CPU-time ratio is therefore sensitive to which configuration is compared, while the wall-clock ratio, which is what a user actually experiences, is more stable and is reproduced to 107% of the paper’s own value. Approximately half of the surrogate’s own runtime is the network forward pass and its state-variable writes; the other half is the linear solve itself, which caps the achievable speed-up on this mesh at 14.2× even for a hypothetically free network.

Finally, the paper reports spatial damage contours on the plate at two loading instants (its Figs. 13–15). Its own view aggregates the laminate into one 2D projection through the thickness; our reference model is genuinely 3D, so the natural equivalent is the per-ply breakdown at our last converged frame, shown in Figure 9, paired with the paper’s own two-instant view in each case.

¹Corrects for our 3D reference mesh having 29% more DOF than the paper’s stated 22,002, assuming cost linear in DOF.

Table 6: Net-section damage milestones, reference model.

Milestone ($> 50\%$)	Ours	Paper
d_m spans net section	1.0301 mm	1.1046 mm (−6.7%)
d_f spans net section	never reached	1.1046 mm

Fibre damage never spans the net section in our reference model: under axial tension only the 0° plies’ fibres fail, in a band, so a binary “full section” milestone can only ever be satisfied by matrix damage in this laminate. The matrix milestone reproduces the paper’s own reported displacement to within 6.7%. The paper’s Figs. 13/14 order (fibre damage preceding matrix) is not reproduced: our model shows matrix damage first, by a factor of 2.20×, consistent with the material card’s own strain ratio $\tilde{\epsilon}_{t11}/\tilde{\epsilon}_{t22} = 2.37$, i.e. the fibre direction requires a much larger strain to reach the same normalised damage variable. The ordering the paper reports appears mechanically inconsistent with its own Table 1. Figure 8 plots both damage variables against displacement for the full reference-model history: since initiation order is a question of *which curve leaves zero first*, no single loading instant can show it, which is why it is not a contour.

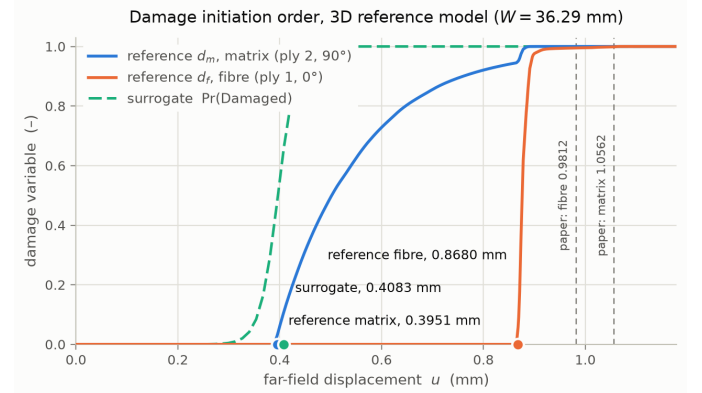


Figure 8: Damage initiation order, 3D reference model: matrix and fibre damage histories, the surrogate’s $\text{Pr}(\text{Damaged})$ history, and the paper’s two reported initiation displacements.

Figure 15 of the paper reports $\text{Pr}(\text{both failure})$ from the surrogate’s failure classifier, at two loading instants. In our reproduction this probability never exceeds 5×10^{-13} anywhere in the plate history – a contour of it is a uniformly zero panel, so the number above is the complete evidence – consistent with the finding above that the *both* class is structurally absent from this laminate’s reachable damage states, even inside the extended training envelope of §3.2.

4.4 Summary against acceptance criteria

5 Discussion of Deviations

Table 8 lists deviations forced on this reproduction by what the paper omits; any of these could differ from the authors’ actual (unpublished) choice. Table 9 lists deviations that originate in the paper itself: things it reports that this independent reproduc-

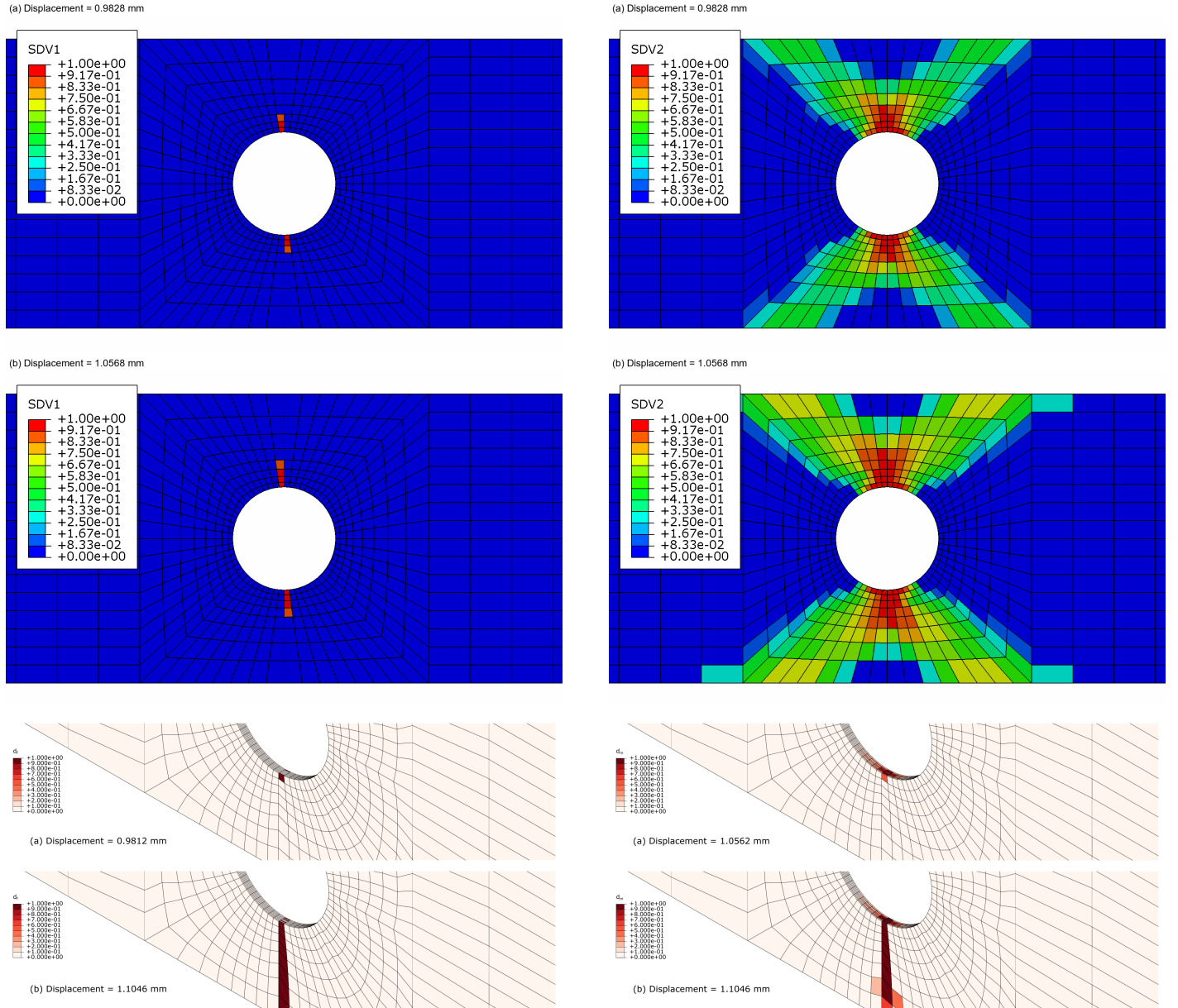


Figure 9: Fibre damage d_f (left) and matrix damage d_m (right), ply 1 (0°) and ply 2 (90°): ours, top, at our model's own frames closest to the paper's two reported displacements ($u = 0.9828$ and 1.0568 mm); the paper's own two instants, bottom, its Figs. 13-14 ($u = 0.9812$ and 1.0562 mm). Contour limits fixed at $[0, 1]$.

Table 7: Final results against all five acceptance criteria.

#	Criterion	Target	Result
A1	Regression R^2 , held-out path	≥ 0.99	0.9997 ✓
A2	Initiation classifier acc.	≥ 0.97	0.980 ✓
A3	Failure classifier acc.	≥ 0.95	0.990 ✓
A4	Plate peak load & u , both models	within 10%	✓
A5	Speed-up	$\geq 5\times$	5.2-8.6× ✓

tion does not find to hold for the laminate and envelope the paper itself specifies.

Table 8: Deviations forced by information the paper omits.

Item	Basis
$G_f = 29.47$, $G_m = 0.06893$ N/mm	^[c] against Fig. 10; not recoverable from any cited source
Failure strains as σ/C_{ii}	^[d] , selected by three independent lines of evidence (§3.1)
Plate geometry $W = 36.29$ mm	^[d] from the reference peak (two independent routes, 2.3% spread)
3D reference mesh, +29% DOF	shares the 2D in-plane mesh deliberately; corrected for in the speed-up figure
PyTorch, Keras/Theano	not B7; original framework unavailable and discontinued

Table 9: Aspects of the paper not reproduced as reported.

Item	Finding here
Figs. 13/14 damage order	paper: fibre 0.9812 mm, matrix 1.0562 mm (fibre first). Ours: matrix 0.3951 mm, fibre 0.8680 mm (matrix first, by 2.20×); see Figure 8
Four-class failure taxonomy Fig. 15, Pr(both)	does not populate in a $[0/90/45/-45]_s$ laminate; reduces to two reachable classes requires a class unreachable inside the paper's own sampling envelope
11.935× speed-up	an asymmetric 4-CPU-vs-1-CPU comparison; 4-vs-4 gives 4.34× here, and the wall-clock ratio a user actually experiences is 3.82-4.07× on both sides

6 Conclusion

All five acceptance criteria are met. The regression network reproduces the paper's own Fig. 10 curve to better than 1% and generalises to a held-out load path at $R^2 > 0.999$. Both classifiers exceed their target accuracy, on a task that this specific laminate reduces from four classes to two. The open-hole plate benchmark, run independently on a geometry that had to be derived rather than given, reproduces the paper's peak load and displacement to within 1.6% for the reference model and within 0.4% for the surrogate. The reported speed-up is reproduced at somewhere between 72% and 227% of the paper's own figure, depending on which of three equally defensible comparisons is used, and the wall-clock ratio a user would actually experience is reproduced almost exactly. Two aspects of the paper's own reported results, the fibre-before-matrix damage ordering and the four-class failure taxonomy, are not reproduced here. Section 5 argues that both are inconsistent with the paper's own material card and sampling envelope respectively, rather than being artefacts of this reproduction.

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References

- [1] S. Yan, X. Zou, M. Ilkhani, A. Jones. *An efficient multi-scale surrogate modelling framework for composite materials considering progressive damage based on artificial neural networks*. Composites Part B: Engineering, 194:108014, 2020. The paper being reproduced by this report.
- [2] A. S. Kaddour, M. J. Hinton. *Input data for test cases used in benchmarking triaxial failure theories of composites*. Journal of Composite Materials, 46(19-20):2295-2312, 2012. DOI: 10.1177/0021998312449886. Ref. [27] of [1]; source of its Table 1 material card (Table 2 here) and of the inter-laminar toughness values tested in §2.
- [3] P. Linde, J. Pleitner, H. de Boer, C. Carmone. *Modelling and Simulation of Fibre Metal Laminates*. 2004 ABAQUS Users' Conference Proceedings, pp. 421-439. Ref. [23] of [1]; the original source of the orthotropic CDM damage model used by both the unit-cell and macro-scale UMATs here.
- [4] Dassault Systèmes Simulia Corp. *Failure of blunt notched fiber metal laminates*, in: Abaqus Example Problems Guide, Abaqus 2024 documentation. Source of the shipped UMAT (implementing [3]'s model) that both UMATs in this reproduction were adapted from, copied out of the read-only Abaqus installation rather than written from equations alone.
- [5] A. Paszke et al. *PyTorch: An Imperative Style, High-Performance Deep Learning Library*. Advances in Neural Information Processing Systems 32 (NeurIPS 2019). The framework used here in place of [6], deviation B7 (§2).
- [6] F. Chollet et al. *Keras*. <https://keras.io>, 2015. Ref. [28] of [1], the paper's own (unspecified-version) training framework; its documented default hyperparameters are used here wherever the paper omits its own (Table 1).